Parenteral Anticoagulants and DOAC Interactions

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Learning Objectives

- By the end of this lecture, providers should be able to perform the following:
  - Identify common and severe drug-drug interactions.
  - Predict the outcome of drug-drug interactions.
  - Determine the clinical significance of a given drug-drug interaction based on scientific literature and patient specific information.
  - Formulate safe and effective drug regimens to manage drug-drug interactions in specific patients.

Anticoagulants

- Heparins
  - Unfractionated Heparin
  - Low Molecular Weight Heparin (LMWH)
- Factor Xa Inhibitors
- Direct Thrombin Inhibitors

All Anticoagulants PD interaction

- All anticoagulants will have a pharmacodynamic interaction with other drugs that have additive effects that <u>increase bleeding risk.</u>
- Agents such as:
  - Thombolytics
  - Other Anticoagulants
  - Antiplatelets (P2Y12 inhibitors)
  - Agents with antiplatelet properties (SSRIs, SNRIs, NSAIDs, Garlic, Ginkgo Biloba, Ginseng, etc)

What are some signs and symptoms of bleeding?

Use of selective serotonin reuptake inhibitors and risk of upper gastrointestinal tract bleeding

- SSRIs ALONE increase the risk of upper gastrointestinal (GI) bleeding **by 3.6-fold**
- Combined use of an SSRI and NSAIDs or an SSRI and low-dose aspirin increased the risk to **12.2 and 5.2 times**, respectively.

1. Dalton S, Johansen C, Mellemkjoer L et al: Use of selective serotonin reuptake inhibitors and risk of upper gastrointestinal tract bleeding. Arch Intern Med 2003; 163:59-64.

DDI Interaction Anticoagulants

- Combination antiplatelet or anticoagulation
  - 21443 elderly acute MI survivors were evaluated for hospitalizations due to bleeding complications given therapy with various AC and AP mono- and combination therapies. There was an increased risk of bleeding (primarily GI) associated with combination therapy, with **aspirin plus a thienopyridine** (eg, clopidogrel) or **aspirin plus warfarin** conferring an approximately 2-fold higher risk than individual agents alone. Triple combination therapy (dual antiplatelets + anticoagulant) resulted in a nearly 3-fold increase.

1. Buresly K, Eisenberg MJ, Zhang X, et al., "Bleeding Complications Associated with Combinations of Aspirin, Thienopyridine Derivatives, and Warfarin in Elderly Patients Following Acute Myocardial Infarction," *Arch Intern Med*, 2005, 165:784-9.

DDI Interaction Anticoagulants

- Warfarin and Aspirin
  - The incidence of bleeding complications was compared in patients treated with warfarin (target INR, 2.2 to 2.8) plus aspirin 150 mg/day (n=1140) to patients treated with warfarin alone (target INR 2.5 to 4.2; n=2026), over a total observation time of 4420 treatment-years. Minor bleeding episodes occurred in 1.4% of the warfarin group and in 2.9% of the warfarin plus aspirin group. <u>The incidence of serious or fatal bleeding episodes did not differ significantly between the two groups</u> (Chan, 1995).

1. Chan TYK: Adverse interactions between warfarin and nonsteroidal antiinflammatory drugs: mechanisms, clinical significance, and avoidance. Ann Pharmacother 1995; 29:1274-1283.

Triple antithrombotic therapy

- A meta analysis compared the risk of bleeding on triple therapy (DAPT + a DOAC) to DAPT alone
  - The HR was 2.34 (95% CI 2.06 to 2.66) for clinically significant bleeding events
    - Little benefit on major cardiovascular events

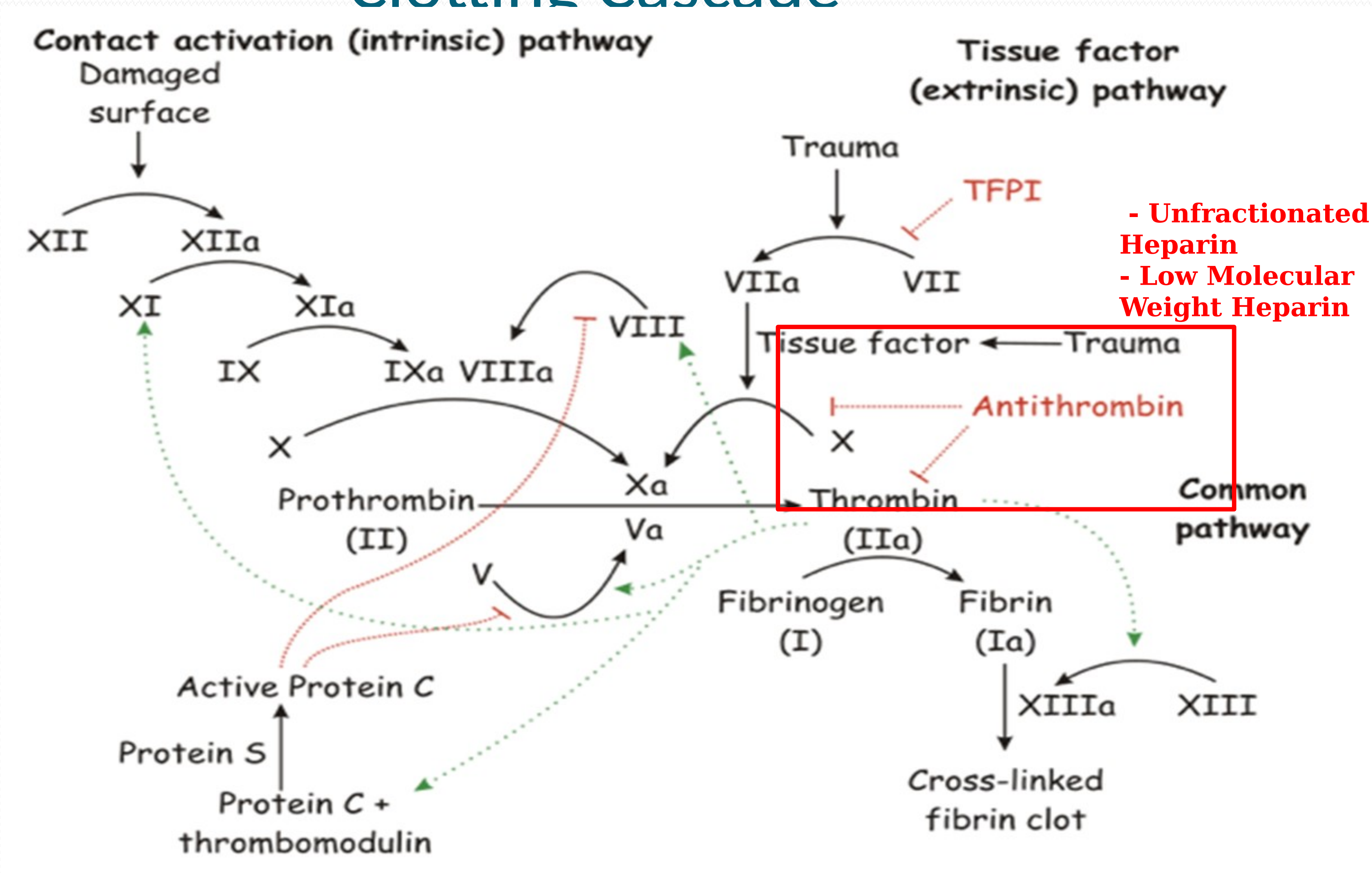
European Heart Journal (2013) 34, 1670–1680

Heparins

Unfractionated Heparin (UFH)

Low Molecular Weight Heparin (LMWH)

Clotting Cascade



- Unfractionated Heparin
- Low Molecular Weight Heparin

1. Image <https://ittakes30.files.wordpress.com/2010/10/clotting-cascade.jpg>

Unfractionated Heparin (UFH)

- MOA: Binds to antithrombin, inhibiting factor Xa and thrombin (factor IIa).
- MW ranges from 3,000-30,000 daltons
- Unpredictable anticoagulant response (variable and extensive binding to plasma proteins and c

1. Drug Monograph [https://www.rxmed.com/b.main/b2.pharmaceutical/b2.1.monographs/CPS-_Monographs/CPS-_(General_Monographs-_H)/HEPARIN.html].
2. Product Information: heparin sodium injection, heparin sodium injection. Baxter Healthcare Corporation, Deerfield, IL, 2006.
3. Image <https://img.medscapestatic.com/pi/features/drugdirectory/octupdate/MPS09421.jpg>

UFH Pharmacokinetics

- Absorption:
  - PO – Avoid (not absorbed)
  - IM – Avoid (hematomas)
  - **IV** - Quick action; continuous infusion preferred
  - **SQ-** onset of action ~20– 30 minutes.
- Distribution: Vd: 0.07 L/kg
  - Protein binding: extensive to low density lipoproteins, globulins (including the alpha-globulin antithrombin III) and to fibrinogen
- Metabolism: Hepatic and reticulo-endothelial system
- Half-life**: 30-90 minutes**
- 2 Elimination Mechanisms:
  - **Enzymatic inaction**: rapid, saturable, zero-order process (lower doses)
  - Renally: Slower, non-saturable, first-order process (higher doses)

1. Product Information: heparin sodium injection, heparin sodium injection. Baxter Healthcare Corporation, Deerfield, IL, 2006.

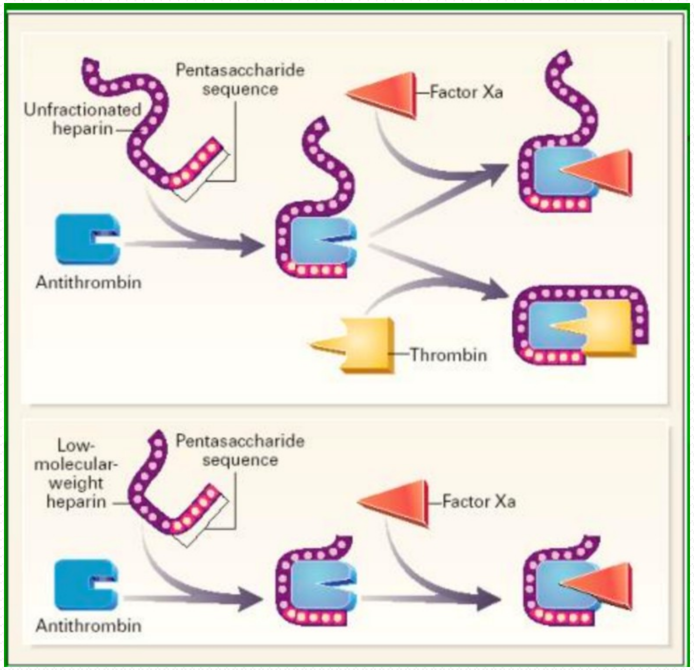
UFH Adverse Effects

- Bleeding
- Thrombocytopenia
  - Platelet count < 100,000
  - Heparin Associated Thrombocytopenia
    - Benign, transient, mild
  - Heparin Induced Thrombocytopenia (HIT)
    - Serious requiring immediate intervention

1. Product Information: heparin sodium injection, heparin sodium injection. Baxter Healthcare Corporation, Deerfield, IL, 2006.

Low Molecular Weight Heparin (LMWH)

- Fragments of UFH produced by chemical of enzymatic depolymizeration.
- ~1/3 of MW of UFH
- Less incidence of HIT
- 3 LMWHs
  - Lovenox® (enoxaparin)
  - Fragmin® (dalteparin)



1. Product Information: Lovenox(R) subcutaneous injection, intravenous injection, enoxaparin sodium subcutaneous injection, intravenous injection. sanofi-aventis US LLC (per FDA), Bridgewater, NJ, 2018.
2. Image <https://image3.slideserve.com/6869041/slide11-n.jpg>.

LMWH Pharmacokinetics

- More predictable response than UFH
- SubQ Bioavailability is ~100%
- Onset of Action: 3-5 hours
- Distribution: Vd 4.3L
- Metabolism: Hepatic via desulfation and depolymerization
- Half-life: **4.5-7 hours**
- Excretion: Predominantly **renally** (<u>decrease dose in CrCL < 30 mL/min</u>)

1. Product Information: Lovenox(R) subcutaneous injection, intravenous injection, enoxaparin sodium subcutaneous injection, intravenous injection. sanofi-aventis US LLC (per FDA), Bridgewater, NJ, 2018.

Drug Interactions with Heparins

- Most drug interactions are due to additive effects with other drugs that can <u>increase bleeding risk.</u>
- Mainly Pharmacodynamic
  - Other Anticoagulants
  - Agents with antiplatelet properties
  - Thrombolytics

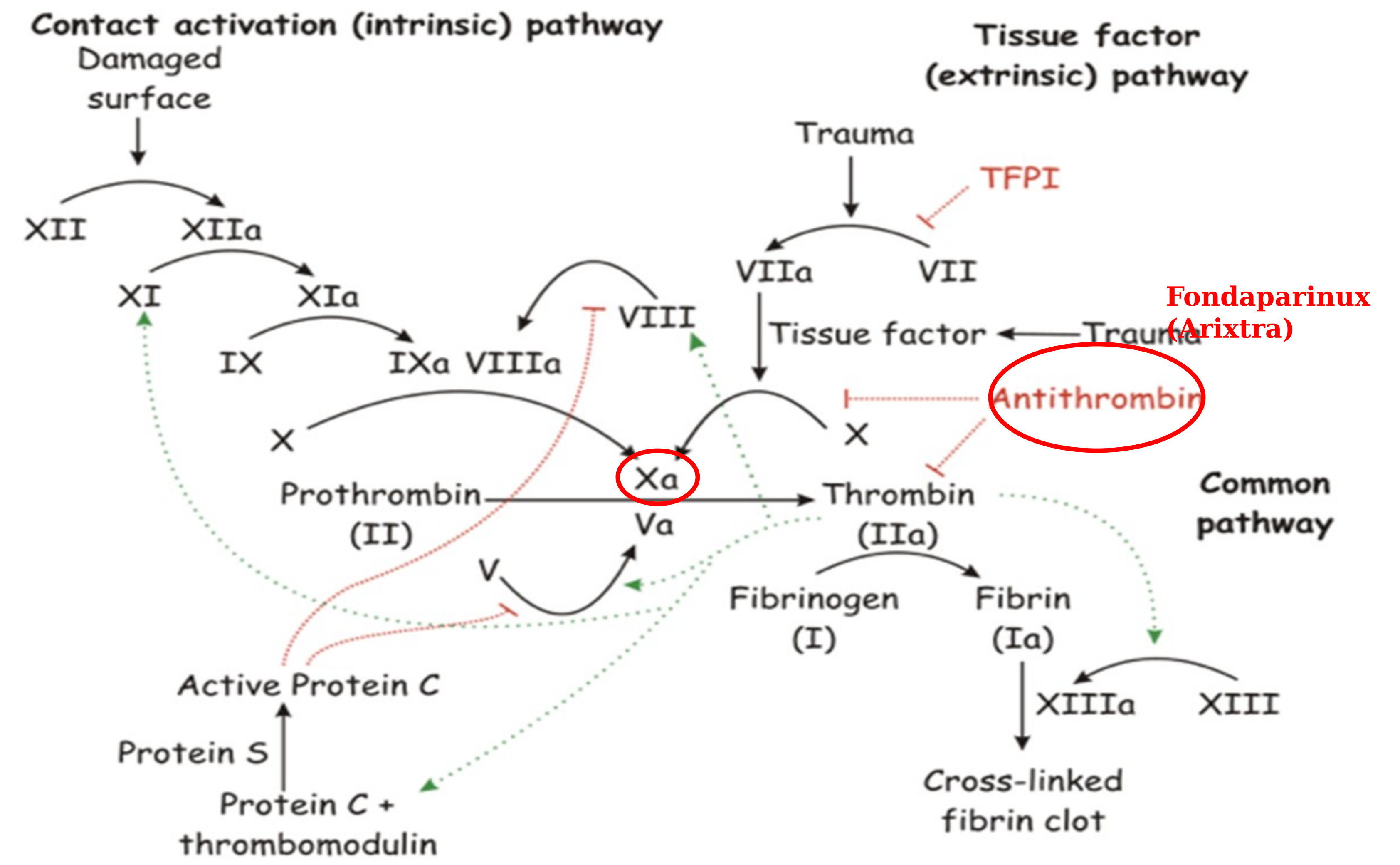
1. Product Information: Lovenox(R) subcutaneous injection, intravenous injection, enoxaparin sodium subcutaneous injection, intravenous injection. sanofi-aventis US LLC (per FDA), Bridgewater, NJ, 2018.

Factor Xa Inhibitors

Factor Xa Inhibitors

- Selectively inhibit factor Xa activity
  - Prevents thrombus generation and clot formation
- Indirect Xa activity
  - Fondaparinux (Arixtra®)
- Direct Xa activity
  - Rivaroxaban (Xarelto®)
  - Apixaban (Eliquis®)
  - Edoxaban (Savaysa®)
  - Betrixaban (Bevyxxa®)

Clotting Cascade



Fondaparinux (Arixtra)

1. Image <https://ittakes30.files.wordpress.com/2010/10/clotting-cascade.jpg>

Fondaparinux (Arixtra)

- MOA: selectively binds to antithrombin, thus inhibiting Factor Xa.
- Used off-label in cases of HIT.

1. Product Information: ARIXTRA(R) subcutaneous injection solution, fondaparinux sodium subcutaneous injection solution. Aspen Global, Inc. (per FDA), Westbury, NY, 2014
2. Image <https://img.medscapestatic.com/pi/features/drugdirectory/octupdate/SDZ06780.jpg>

Fondaparinux Pharmacokinetics

- Absorption: SubQ Bioavailability ~100%.
- Time to peak = 3 hours
- Distribution: Vd: 7 – 11 L
  - Protein binding 94%
- Excretion: Renal: 77% unchanged
- Half life: 17 – 21 hours

1. Product Information: ARIXTRA(R) subcutaneous injection solution, fondaparinux sodium subcutaneous injection solution. Aspen Global, Inc. (per FDA), Westbury, NY, 2014

Fondaparinux Drug Interactions

- Like the heparins, most interactions are due to additive bleedings with other drugs that can increase bleeding risk.
- Mainly Pharmacodynamic
  - Other Anticoagulants
  - Agents with antiplatelet properties
  - Thrombolytics

1. Product Information: ARIXTRA(R) subcutaneous injection solution, fondaparinux sodium subcutaneous injection solution. Aspen Global, Inc. (per FDA), Westbury, NY, 2014

Direct-Acting Oral Anticoagulants (DOACs)

DOACs

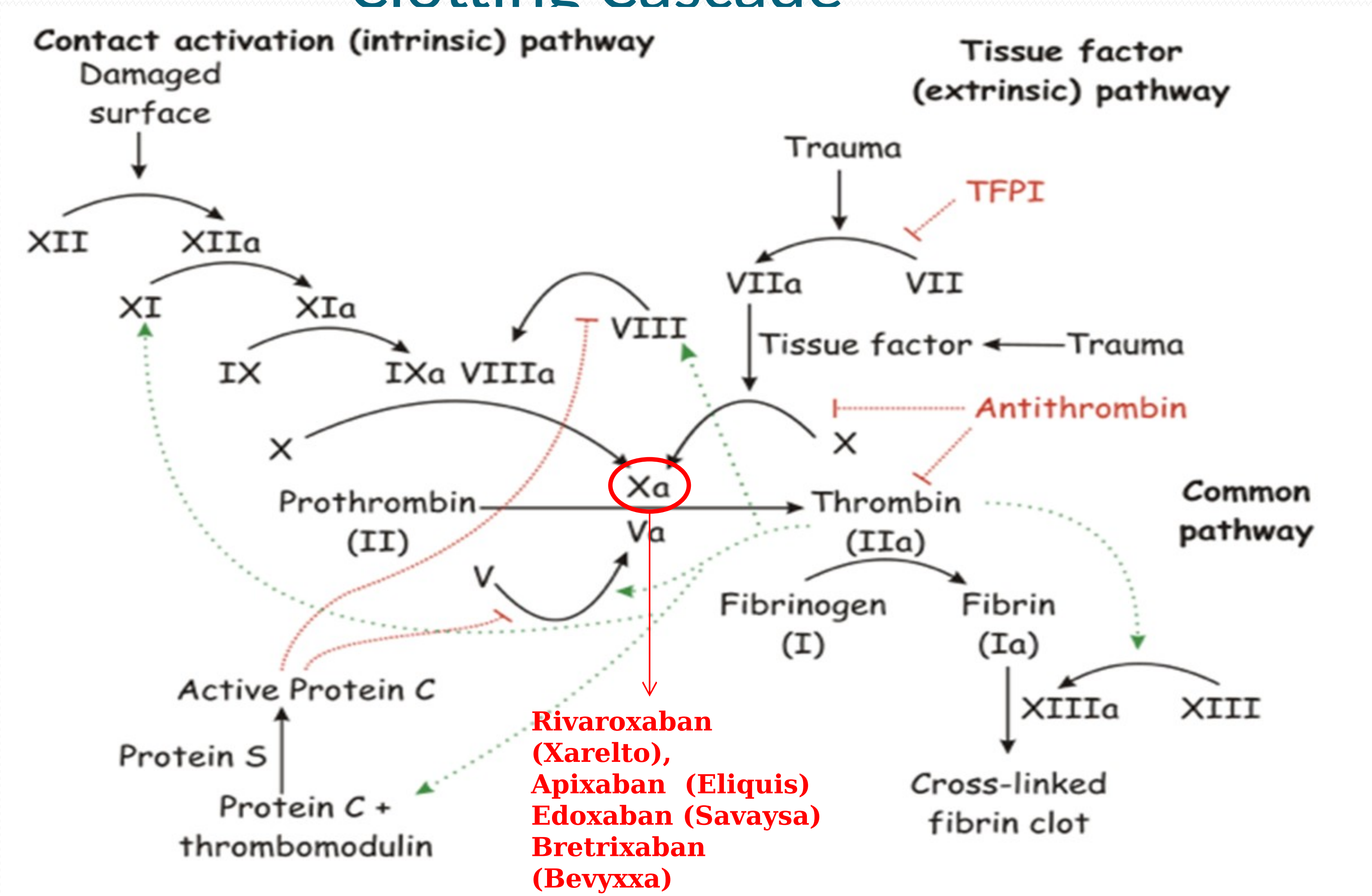
- Direct-Acting Oral Anticoagulants (DOAC)
- They are a class of newer oral blood thinners that may be more convenient than warfarin.
- Consists of:
  - Direct Factor Xa Inhibitors
    - Rivaroxaban (Xarelto®)
    - Apixaban (Eliquis®)
    - Edoxaban (Savaysa®)
    - Betrixaban (Bevyxxa®)
  - Direct Thrombin Inhibitor
    - Dabigatran (Pradaxa®)

Direct Factor Xa Inhibitors

- MOA: selectively inhibitors Factor Xa without the need of a cofactor.
- All available orally.
- Must consider renal dosing.

1. Husted S, de Caterina R, Andreotti F, et al. Non-vitamin K antagonist oral anticoagulants (NOACs): No longer new or novel. Thromb Haemost 2014; 111:781

Clotting Cascade



Rivaroxaban (Xarelto),
Apixaban (Eliquis)
Edoxaban (Savaysa)
Bretrixaban (Bevyxxa)

Rivaroxaban (Xarelto®)

- Indications:
  - Non-valvular A-fib
  - Treatment of DVT/PE
  - Prophylaxis of DVT/PE
- Take doses > 15 mg with FOOD (to increase bioavailability).

1. Product Information: XARELTO(R) oral tablets, rivaroxaban oral tablets. Janssen Pharmaceuticals Inc (per FDA), Titusville, NJ, 2018.
2. Image: <https://www.empr.com/wp-content/uploads/sites/7/2018/12/cc125257f1d5491abf848c82b05ce527-xarelto20mgmono.jpg>

Rivaroxaban Pharmacokinetics

- Absorption: oral bioavailability 66 – 100%; dose-dependent
  - Food increases bioavailability by 39% (15 and 20 mg)
  - Time to peak = 2-4 hours
- Distribution: Vd = 50 L
  - Protein binding, albumin: 92% - 95%
- Metabolism: Liver
  - Primary site via Cyp3A4/5, CyP2J2, and hydrolysis
- Excretion: Renally ~66%, feces 28%
- Half-life elimination: 5-9 hours

1. Product Information: XARELTO(R) oral tablets, rivaroxaban oral tablets. Janssen Pharmaceuticals Inc (per FDA), Titusville, NJ, 2018.

Rivaroxaban Drug Interactions

- Pharmacodynamic interactions:
  - Drugs with additive effects that increase bleeding risk
- Pharmacokinetic interactions: Major CYP3A4 substrate and P-GP

What are some common CYP3A4 Inducers?

What are some common CYP3A4 Inhibitors?

Rivaroxaban PK Interactions

- Strong CYP3A4 and P-gp Inducers
  - Carbamazepine
  - Phenytoin
  - Rifampin
  - St John’s Wort
- Strong CYP3A4 and P-gp Inhibitors
  - Ketoconazole, Itraconazole
  - Lopinavir/ritonavir
  - Indinavir
  - Conivaptan
- For moderate inhibitors: Benefit must outweigh risk in pts CrCL 15-80 mL/min

Apixaban (Eliquis®)

- Indications
  - Non-valvular A-fib
  - Treatment of DVT/PE
  - Prophylaxis of DVT /PE
  - Lower dose in A-fib if patient has <u>at least 2 of the following</u>:
    - Age $\geq$ 80 years
    - Weight $\leq$ 60 kg
    - SCr $\geq$ 1.5 mg/dL

1. Product Information: ELIQUIS(R) oral tablets , apixaban oral tablets. Bristol-Myers Squibb Company and Pfizer Inc (per FDA), Princeton, NJ, 2015.
2. Image <rxzone.net/wp-content/uploads/2018/07/322229.jpg>

Apixaban Pharmacokinetics

- Absorption: Bioavailability 50%
  - Time to peak: 3-4 hours
- Distribution: Vd = 21 - 61 L
  - Protein binding: 87%
- Metabolism: Hepatic
  - Mainly via CYP3A4
  - Substrate pf CyP3A4 and P-gp
- Excretion: Renally (~27%), feces (majority)
- Half-life: 11-12 hours (dose-dependent)

1. Product Information: ELIQUIS(R) oral tablets , apixaban oral tablets. Bristol-Myers Squibb Company and Pfizer Inc (per FDA), Princeton, NJ, 2015.

Apixaban Drug Interactions

- Pharmacodynamic interactions:
  - Drugs with additive effects that increase bleeding risk
- Pharmacokinetic interactions: Major CYP3A4 substrate and P-gp
- For those receiving Eliquis 5 mg or 10 mg, the dose <u>should be decreased by 50%, </u>when co-administered with drugs that are both P-gp and strong CYP3A4 inhibitors.

1. Product Information: ELIQUIS(R) oral tablets , apixaban oral tablets. Bristol-Myers Squibb Company and Pfizer Inc (per FDA), Princeton, NJ, 2015.

Apixaban PK Interactions

- Strong CYP3A4 and P-gp Inducers
  - Avoid concomitant use of apixaban with combined P-gp and Strong CYP3A4 inducers because such drugs will decrease exposure to apixaban.
- For patients taking 2.5 mg BID, avoid strong dual inhibitors.

1. Product Information: ELIQUIS(R) oral tablets , apixaban oral tablets. Bristol-Myers Squibb Company and Pfizer Inc (per FDA), Princeton, NJ, 2015.

Edoxaban (Savaysa®)

- Indications:
  - Non-valvular A-fib
  - Treatment of DVT/PE
- Not recommended in CrCL < 15 mL/min and CrCL > 95 mL/min.

1. Product Information: SAVAYSA(TM) oral tablets, edoxaban oral tablets. Daiichi Sankyo, Inc. (per FDA), Parsippany, NJ, 2015.
2. Image: <https://www.prestoimages.net/store20/rd648/648_pd2227643_1.png>

Edoxaban

- Absorption: Oral Bioavailability: 63-72%
  - Time to peak 1-2 hours
- Distribution: Vd: 107 L
  - Protein binding: 55%
- Metabolism: Minimal hepatic
  - Substrate of P-gp, CyP3A4
  - Weak inhibitor of P-gp, OATP1B1, and OATP1B3
- Excretion: Renal ~50%
  - Half-life = 11.5 hours

1. Product Information: SAVAYSA(TM) oral tablets, edoxaban oral tablets. Daiichi Sankyo, Inc. (per FDA), Parsippany, NJ, 2015.

Edoxaban Drug Interactions

- Pharmacodynamic interactions:
  - Drugs with additive effects that increase bleeding risk
- Pharmacokinetic interactions: Substrate of Pgp and CYP3A4

1. Product Information: SAVAYSA(TM) oral tablets, edoxaban oral tablets. Daiichi Sankyo, Inc. (per FDA), Parsippany, NJ, 2015.

Edoxaban PK Interactions

- Avoid use with rifampin due to induction of p-gp-medicated edoxaban efflux transport.
- No dose reduction in patients concomitantly receiving P-gp inhibitors.

1. Product Information: SAVAYSA(TM) oral tablets, edoxaban oral tablets. Daiichi Sankyo, Inc. (per FDA), Parsippany, NJ, 2015.

Betrixaban (Bevyxxa®)

- Indications:
  - Prophylaxis of DVT (hospital and extended) in hospitalized adults

1. Product Information: BEVYXXA(TM) oral capsules, betrixaban oral capsules. Portola Pharmaceuticals Inc (per FDA), South San Francisco, CA, 2017.

Betrixaban Pharmacokinetics

- Absorption: Oral bioavailability: 34%
  - Time to peak: 3-4 hours
- Distribution: Vd: 32 L/kg
  - Protein binding: 60%
- Metabolism: Substrate of P-gp
- Excretion: Fecal 85%; Renal: ~11%
  - Half-life: 19-27 hours

1. Product Information: BEVYXXA(TM) oral capsules, betrixaban oral capsules. Portola Pharmaceuticals Inc (per FDA), South San Francisco, CA, 2017.

Betrixaban Drug Interactions

- Pharmacodynamic interactions:
  - Drugs with additive effects that increase bleeding risk
- Pharmacokinetic interactions:
  - Reduce the dose of betrixaban for patients receiving or starting concomitant P-gp inhibitors
  - Avoid betrixaban in patients with severe renal impairment (CrCl < 30 mL/min) receiving P-gp inhibitors.

1. Product Information: BEVYXXA(TM) oral capsules, betrixaban oral capsules. Portola Pharmaceuticals Inc (per FDA), South San Francisco, CA, 2017.

Direct Thrombin Inhibitor

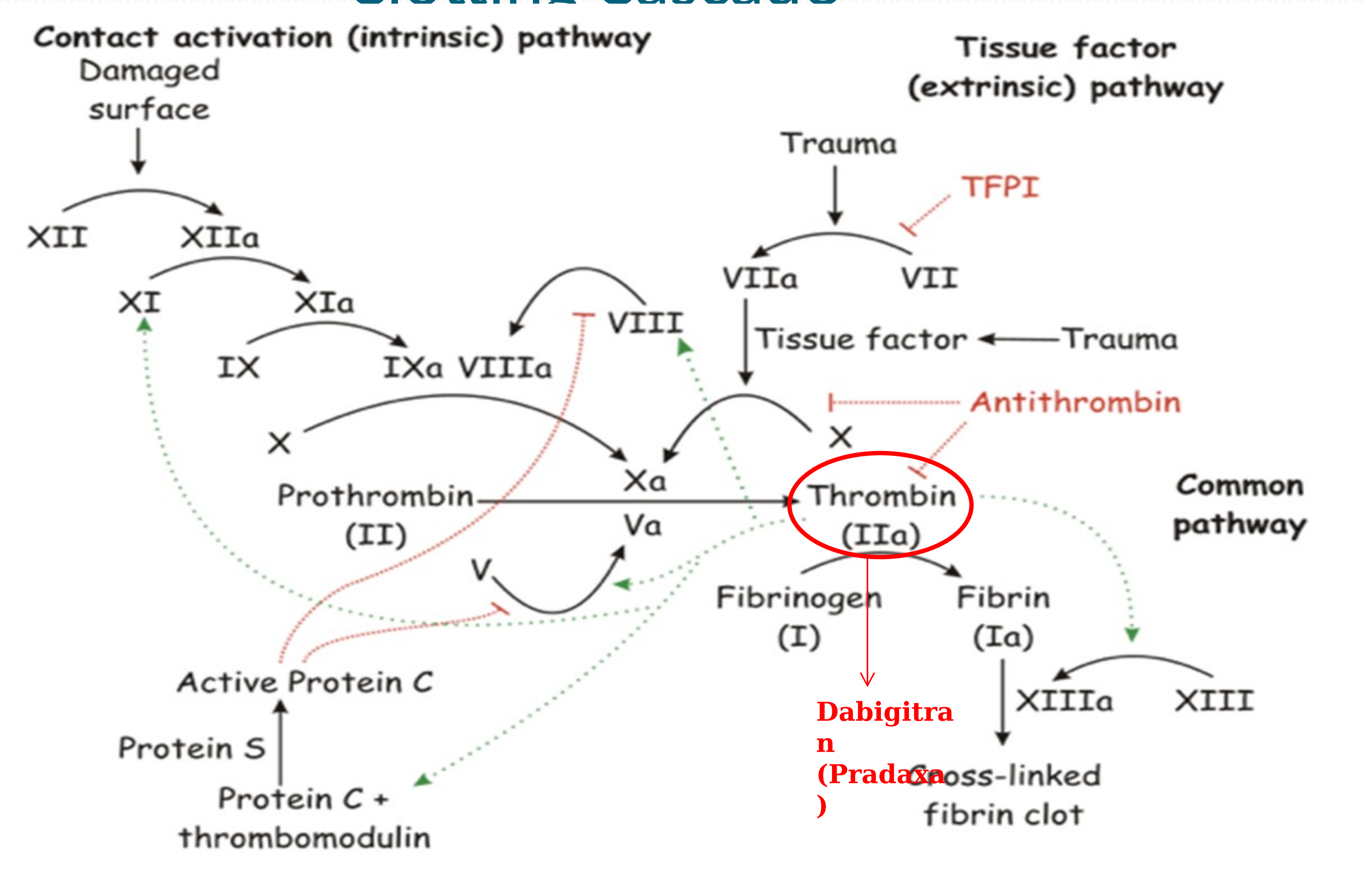
Dabigatran (Pradaxa®)

Dabigatran (Pradaxa®)

- MOA: directly inhibits thrombin.
- Indications:
  - Non-valvular A-fib
  - Treatment of DVT/PE
  - Prophylaxis of DVT/PE
- Dispense in original package and use within 4 months
- GI Symptoms > 10% of patients

1. Product Information: PRADAXA(R) oral capsules, dabigatran etexilate mesylate oral capsules. Boehringer Ingelheim Pharmaceuticals, Inc. (per manufacturer), Ridgefield , CT, 2014.
2. Image <https://mdedge-files-live.s3.us-east-2.amazonaws.com/files/s3fs-public/images/articles/hemtimes/dabigatran-pradaxa-BI-220.jpg>

Clotting Cascade



1. Image < https://ittakes30.files.wordpress.com/2010/10/clotting-cascade.jpg>.

Dabigatran Pharmacokinetics

- Absorption: Oral bioavailability: 3 – 7%
  - Time to peak: 1-2 hours
- Distribution: Vd = 50-70 L
  - Protein binding: 35%
- Metabolism: Hepatic
  - Hydrolyzed to active form by plasma and hepatic esterases
- Excretion: Renal = 80%
  - Half-life: 12-17 hours

1. Product Information: PRADAXA(R) oral capsules, dabigatran etexilate mesylate oral capsules. Boehringer Ingelheim Pharmaceuticals, Inc. (per manufacturer), Ridgefield , CT, 2014.

Dabigatran Drug Interactions

- Pharmacodynamic interactions:
  - Drugs with additive effects that increase bleeding risk
- Pharmacokinetic interactions: Substrate of P-gp
  - Avoid co-administration with p-gp inducer rifampin.
  - In patients with moderate renal impairment (CrCl 30-50 mL/min) on concomitant P-gp inhibitors, consider reducing dabigatran dose to 75 mg twice daily.
  - P-gp inhibitors with severe renal impairment

1. Product Information: PRADAXA(R) oral capsules, dabigatran etexilate mesylate oral capsules. Boehringer Ingelheim Pharmaceuticals, Inc. (per manufacturer), Ridgefield , CT, 2014

Patient Case

- JM is a 54 y/o woman who has been taking Xarelto 15 mg daily for atrial fibrillation and amitriptyline 50 mg QHS for migraine prophylaxis. Shortly after the death of her husband, she has experienced depression. Her prescriber then started her on citalopram 40 mg daily.

What is the patient at risk for due to the drug combinations?

Increased bleeding risk but decreased risk of arrhythmia

Increased clotting risk but decreased risk of bleeding

Increased risk of bleeding and increased risk of arrhthymia

Increased risk of worsening depression

Worsening depression and increased clotting risk

Which of the following agents would NOT increase her bleeding risk and can be trialed?

Bupropion	**A**
Effexor	**B**
Pristiq	**C**
Lexapro	**D**

Summary

- All anticoagulants have a pharmacodynamic interaction with agents that increase bleeding risk.
- Important to consider agent specific PK interactions.
- Interaction management can be determined through analysis of bleeding risk versus embolic risk.

References

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- Product Information: SAVAYSA(TM) oral tablets, edoxaban oral tablets. Daiichi Sankyo, Inc. (per FDA), Parsippany, NJ, 2015.
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Questions?

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